

Cluster-Based Student Profiling and Explainable Early Engagement Prediction Using the Open University Learning Analytics Dataset (OULAD)

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Abstract — This paper examines whether cluster-derived behavioural profiles can improve early, interpretable at-risk prediction for distance learners in the Open University Learning Analytics Dataset (OULAD). The pipeline combines K-means clustering on VLE clickstream volume and regularity, XGBoost for withdrawal prediction, and SHAP for archetype-level explanation. On the FFF module (2013 train, 2014 test), clustering identifies four archetypes — High Engaged, Moderate, Disengaged, and No VLE Record — whose withdrawal rates diverge sharply, from 1.3% to 95.0%. Against a strong baseline (macro-F1 = 0.8140, AUC-ROC = 0.9085), adding the cluster label raises macro-F1 by up to 0.0070 and AUC-ROC by up to 0.0076, though the gain shrinks with richer baselines and does not reach statistical significance. SHAP analysis shows the label’s contribution differs systematically by archetype, pushing toward withdrawal for students with no VLE activity and away from it for engaged archetypes. Two robustness checks qualify these findings: transfer to the BBB module without retraining loses substantial performance (macro-F1 –21.8pp, AUC-ROC –30.7pp), and a multi-class extension shows the cluster label’s advantages do not survive a harder four-outcome task. Overall, cluster-aware modelling offers a modest but consistent directional improvement in prediction and interpretability within a single, previously-seen module, motivating future work on cross-module adaptation before deployment.

Keywords — learning analytics, OULAD, student profiling, at-risk prediction, explainable AI, K-means clustering, XGBoost, SHAP, cross-module transferability

I INTRODUCTION

The Open University Learning Analytics Dataset (OULAD) records 32,593 student module registrations across seven courses delivered in 2013-2014, which provides Virtual Learning Environment (VLE) clickstream logs, demographic attributes, and assessment records. As a widely adopted benchmark in learning analytics research, OULAD offers a rich basis for studying student disengagement in distance education (Kuzilek et al. 2017).

However, approximately 32% of those students received a final outcome of Withdrawn, which indicates a failure to complete the module. Furthermore, a large proportion of these withdrawals occur during the early stages of a student’s first module, before any formal assessment signal can alert institutional staff, leaving a critically narrow window for timely intervention. To address this early disengagement problem, this project aims to construct interpretable student behavioural profiles via k-Means clustering and develop an explainable machine learning model for early prediction of at-risk students.

A Research Questions

This study addresses three research questions:

1. Which VLE behavioural features most effectively separate learners into meaningful and stable engagement profiles on OULAD?
2. Does adding cluster-derived profile membership improve early at-risk prediction performance (macro-F1 and AUC-ROC) compared with strong raw-feature baselines?
3. Do SHAP explanations differ systematically across learner profiles, and which behavioural drivers by profile contribute most to withdrawal risk?

The remainder of this paper is organised as follows. Section II reviews prior work on learning analytics, student profiling, at-risk prediction, and explainable AI in educational contexts. Section III describes the proposed pipeline in detail, covering data preprocessing, K-means clustering, feature engineering, XGBoost modelling, and SHAP-based explanation. Section IV presents and evaluates the experimental results, including cluster characterisation, prediction performance, SHAP analysis, and robustness checks. Finally, Section V concludes the paper, summarises the key findings, discusses limitations, and outlines directions for future work.

II RELATED WORK

A Dataset Overview

OULAD anonymises the actual course titles, which identifies all seven modules by three-letter codes (AAA–GGG). Based on the disciplinary classifications reported in published OULAD studies (Kuzilek et al. 2017, Alhakbani & Alnassar 2022), AAA, BBB, and GGG are Social Science courses, while CCC, DDD, EEE, and FFF are STEM courses. Each module is offered in one or two annual presentations coded as $\mathcal{J}$ (January start) or $\mathcal{B}$ (February start), with presentation lengths ranging from 234 to 269 days. This project uses FFF (STEM) as the primary analytical module and BBB (Social Science) as a cross-domain robustness check; The reason is that the project prediction target is *final_result*, with *Withdrawn* treated as the at-risk class and the classes are imbalanced and the behavioural data are sparse early in the course. Therefore, the dataset is well suited to both clustering-based profiling and early-warning prediction.

Several publicly available learning analytics datasets were considered before selecting OULAD; Table 1 summarises the key properties of each.

Table 1: Comparison of candidate datasets for at-risk prediction research.

Dataset	Scale	Clickstream	Assessments	Withdrawal label
HarvardX–MITx MOOC (Ho et al. 2014)	476,532	coarse counts	none	certification only
KDD Cup 2010 (Baker et al. 2010)	574,239	step-level	tutor-based	no
OULAD (Kuzilek et al. 2017)	32,593	timestamped	multi-type	yes

The HarvardX–MITx MOOC dataset (Ho et al. 2014), though very large (476,532 registrations across 17 edX courses), provides only coarse activity counts, lacks formal graded-assessment schedules, and does not record re-sit attempts; in addition, MOOC withdrawal dynamics differ from structured distance higher education, and the data do not support fine-grained weekly behavioural engineering for this early-warning task. The KDD Cup 2010 dataset (Baker et al. 2010) offers rich problem-step logs but is limited to a single secondary-school mathematics course, lacks demographic/registration metadata, and has no withdrawal or completion label, so it is not suitable for multi-module, multi-demographic at-risk prediction. By contrast, OULAD (Kuzilek et al. 2017) is selected because it uniquely combines timestamped VLE clickstream logs, demographic/background attributes, graded assessments (dates and weights), and explicit final-result labels including *Withdrawn*, across seven modules in both STEM and Social Science domains and two presentation years (2013–2014). This combination directly supports interpretable K-means profiling plus explainable early-warning modelling, and its benchmark status (Alhakbani & Alnassar 2022, Jin et al. 2024) enables direct comparison with prior work in Table 1.

This project builds its analytical dataset from the raw OULAD release, which is organised into seven linked tables: *courses* describes the length of each module presentation, *assessments* records the type, date, and weight of each assessment, *vle* maps site IDs to activity types, *studentInfo* contains demographic and background variables together with the final result, *studentRegistration* tracks registration and unregistration timing, *studentAssessment* stores assessment submissions and scores, and *studentVle* captures timestamped clickstream events on the virtual learning environment.

The natural unit of analysis is a student-module-presentation instance. In practice, the tables are joined on `code_module` and `code_presentation`, with `id_student` used to align the student-level records and `id_site` used to enrich VLE events with activity metadata. The resulting student traces are cleaned and aggregated into weekly behavioural features, so that each row in the analytic dataset summarises one learner’s registration, background, assessment, and platform-usage profile over time.

B Unsupervised Learner Profiling

Clustering-based learner profiling builds on established engagement and attrition research. Recent learning-analytics syntheses treat engagement as multidimensional, but show behavioural participation indicators — time on task, resource-access frequency, and sustained week-level activity — as the strongest operational predictors of persistence and completion (Fredricks et al. 2021, Hew et al. 2023). In distance education, VLE clickstream traces are therefore a practical proxy for engagement and naturally motivate clustering-based learner profiling.

Across datasets and methods, clustering studies repeatedly recover a small interpretable profile set (typically three to five groups): *highly engaged*, *moderately engaged*, *disengaged*, and, in distance-learning settings, *no recorded VLE activity*. The no-activity group is consistently the highest-risk group (Alhakbani & Alnassar 2022, Jin et al. 2024, El Ghali et al. 2025, Mdaghri-Alaoui et al. 2025). Post-2020 evidence supports the same gradient: sustained participation aligns with completion, while near-zero activity aligns with attrition. Hew et al. (2023), analysing 35 large-scale online courses, likewise identifies engagement frequency as the strongest completion predictor. In OULAD, this archetype–outcome pattern appears in both STEM and Social Science modules, suggesting a domain-general engagement–withdrawal relationship.

Beyond clustering, early-warning prediction spans several model families. Rule-based systems are transparent but threshold-dependent and weak on non-linearity (Alhakbani & Alnassar 2022); regression/logistic models are interpretable but constrained by linearity and sparse, high-dimensional weekly VLE matrices (Kabathova & Drlik 2021); and deep models (LSTM/GRU) capture temporal patterns but are costly and less interpretable. On OULAD, deep models offer only marginal accuracy gains over tree ensembles in some settings (Waheed et al. 2022, Mansouri et al. 2026). XGBoost is a practical middle ground because it handles sparse, imbalanced tabular data, models non-linear interactions, scales well, and remains SHAP-compatible (Jin et al. 2024). A key open question is whether cluster-derived profile membership adds predictive signal beyond the raw variables used to construct those profiles.

OULAD-focused reviews also identify profiling as a dominant analytic pattern (Alhakbani & Alnassar 2022, Jin et al. 2024). Representative studies confirm this but leave important gaps: Alzahrani et al. (2025) report valid K-means profiles on a single course (127 students) without prediction; El Ghali et al. (2025) report four OULAD profiles (silhouette 0.93) after PCA, but mix demographics, outcomes, and engagement traces and do not extend to predictive or cluster-level explanation analysis; Mdaghri-Alaoui et al. (2025) distinguish behavioural and demographic profile pipelines with very high classifier accuracy (>0.99), yet use coarse behavioural inputs (mainly average score and total clicks). Overall, prior work consistently finds risk-differentiated archetypes but rarely unifies profile discovery, downstream supervised prediction, early-warning utility, and cluster-level explanation.

C Supervised Engagement Prediction

Al-Shabandar et al. (2019) built a cross-dataset supervised benchmark using Harvard MOOC (HarvardX learner data) and OULAD records, comparing Gradient Boosting Machine (GBM) against neural-network models for learning-achievement prediction. Using full-course VLE clickstream aggregates as model inputs, GBM achieved 95.2% and outperformed the neural alternatives in their setting, positioning tree-ensemble methods as a strong baseline for VLE-based prediction; however, the setup relied on full-course aggregated signals and did not provide an early-stage warning analysis or model-level explainability.

Torkhani & Rezgui (2025) subsequently benchmarked six machine learning (ML) classifiers and four deep learning models on the OULAD AAA module (748 students); the target was student performance prediction, with Random Forest leading ML performance (accuracy 82.73%) and long short-term memory (LSTM) leading overall (83.41%); yet the study covers one module with no clustering or explainability.

AlFaress et al. (2025) evaluated five classifiers against a 44-feature set of VLE clickstream, assessment, and demographic variables on OULAD; Gradient Boosting achieved the highest performance (accuracy 91.02%, F1-score 91.18%), and the prominence of the *imd_band* deprivation index as a top predictor flags an algorithmic-bias risk if deployed without a fairness audit.

Alalawi et al. (2025) validates a prediction-intervention pipeline on an institutional tertiary-education dataset in a live setting (pass rates +35%), but relies solely on assessment scores without VLE traces or clustering.

Most recently, Mansouri et al. (2026) achieved 90% accuracy on OULAD using a synthetic minority over-sampling technique (SMOTE)-genetic algorithm (GA)-gated recurrent unit (GRU) deep learning pipeline; how-

ever, the authors themselves note that GRU predictions are difficult to explain, motivating the use of interpretable tree-based models in the present project. Taken together, these supervised studies demonstrate that strong headline accuracy is achievable, yet most models treat learners as a homogeneous population and remain weakly connected to behaviour-derived profiles, early-warning timing constraints, and cluster-specific interpretability.

D Explainability in Educational ML

Explainability techniques have become essential for translating black-box model outputs into actionable insights for educators and policymakers. Ahmed et al. (2025) used generic Kaggle educational datasets, applied SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to an ensemble voting regressor, and reported that prior academic scores accounted for approximately 62% of global SHAP influence while study hours ranked second. However, the datasets contain no longitudinal VLE clickstream data or learner-clustering structure, the study cannot reveal how feature importances vary across engagement archetypes.

Islam et al. (2025) used a Portuguese higher-education dropout dataset, applied SHAP, LIME, and Explain Like I'm 5 (ELI5) to an Extreme Gradient Boosting (XGBoost) model, and identified curricular progression as the dominant predictor. However, the analysis remained global, with no VLE traces or cluster-stratified explanation.

Beyond post-hoc explanation, a small but growing body of work has closed the loop between interpretable predictions and actual student-outcome improvement through targeted intervention. Herodotou et al. (2020) conducted a large-scale implementation study of the OU Analyse predictive system at the Open University — the same institution that produced the OULAD dataset used in the present work — and examined how tutors responded to at-risk dashboard alerts across multiple module presentations; tutors who engaged with the risk indicators and contacted flagged students within the first four weeks reported significantly higher student retention compared with module presentations where alerts were ignored, providing direct institutional evidence that actionable, interpretable risk signals translate into measurable academic benefit.

Liu et al. (2023) subsequently applied SHAP-explained gradient boosting to a Chinese university LMS dataset, surfaced weekly VLE inactivity and missed formative assessments as the dominant risk drivers, and embedded the explanations in an adviser dashboard; a follow-up evaluation across two semesters showed that students flagged and contacted by advisers had a retention rate 12 percentage points higher than uncontacted at-risk peers, directly linking model interpretability to intervention effectiveness.

Hlosta et al. (2022) integrated an OULAD-trained early-warning classifier into the Open University's live analytics dashboard (OU Analyse); tutors who acted on the dashboard alerts within the first four weeks of a module achieved statistically significant reductions in withdrawal rates compared with non-acting tutors, confirming that even modest improvements in model explainability and timeliness can produce real gains in student retention within the distance- education context most relevant to the present study. Collectively, these intervention studies establish that explainability is not merely an academic exercise: interpretable risk signals, when communicated clearly to educators, enable timely and targeted support that demonstrably improves student performance and reduces dropout.

E Research Gap

The absence of an integrated clustering-prediction-explainability pipeline has been noted across nearly a decade of learning analytics research: Alhakbani & Alnassar (2022) surveyed 18 OULAD benchmark studies up to 2022, and Jin et al. (2024) subsequently reviewed a further 17 OULAD papers spanning 2017–2024; both found model explanation consistently absent. The studies reviewed above confirm this pattern persists into 2025: clustering studies construct behavioural profiles but stop short of feeding them into a downstream supervised model; prediction studies achieve strong accuracy but treat the student population as a homogeneous mass; and explainability studies operate at population level rather than aligned with cluster-aware pipelines. The present project directly addresses this compound gap by integrating K-means clustering on VLE interaction sequences with XGBoost-based early-warning prediction and per-cluster SHAP explainability: concretely, to the best of our knowledge based on the reviewed literature, no prior OULAD study has simultaneously derived behavioural learner clusters from VLE interaction traces, evaluated early-warning at-risk prediction performance,

and reported cluster-specific model explanations in a single end-to-end pipeline. Table E summarises the key properties and limitations of the most representative prior studies reviewed in this section. This compound gap maps directly onto the three research questions stated in Section A: the absence of cluster-aware behavioural profiling motivates RQ1 (which VLE features best separate learners into stable engagement profiles?); the lack of any study that feeds cluster labels into a downstream predictor motivates RQ2 (does profile membership improve early at-risk prediction?); and the universal reliance on population-level rather than cluster-stratified explanation motivates RQ3 (do SHAP explanations differ systematically across learner profiles, and which profile-specific drivers contribute most to withdrawal risk?).

Table 2. : Comparison of representative studies on student performance and at-risk prediction.

Study	Datasets	Prediction target	Input data/features	Best reported result	Main limitation
Al-Shabandar et al. (2019)	Harvard MOOC + OULAD	Learning achievement	Full-course VLE clickstream aggregates	GBM: 95.2%	No clustering or explainability layer
Torkhani and Rezgui (2025)	OULAD (AAA module)	Student performance	ML/DL benchmark features (748 students)	LSTM: 83.41% (RF: 82.73%)	Single module only; no clustering or explainability
Alfaress et al. (2025)	OULAD	At-risk / outcome prediction	44 features from VLE clickstream, assessment, and demographics	Gradient Boosting: 91.02% accuracy, 91.18% F1	Potential fairness risk from socioeconomic predictor dominance
Alalawi et al. (2025)	Institutional HE dataset	Performance-risk intervention	Assessment-score pipeline in live tertiary deployment	Pass-rate improvement: +35%	No VLE traces or clustering
Mansouri et al. (2026)	OULAD	Student performance	SMOTE-GA-GRU pipeline	Accuracy: 90%	Limited explainability of GRU predictions

III SOLUTION

This section presents an end-to-end early-warning pipeline for OULAD that combines supervised prediction, unsupervised profiling, and post-hoc explainability under a deployment-oriented evaluation design. As summarised in Table 2, the six tiers run sequentially: B1 builds a raw-feature XGBoost baseline; B2 engineers VLE features and derives K-means profiles to answer RQ1; I1 adds profile identity to B1 and compares macro-F1/AUC-ROC against the baseline to answer RQ2; I2 applies SHAP for comparative interpretation to answer RQ3; A1 tests cross-module transferability; and A2 tests robustness under a multi-class outcome setting.

Table 3: Project goals by tier and research-question coverage.

Tier	Tier ID	Description	RQ Addressed
Basic	B1	Train raw-feature XGBoost baseline on OULAD; report Accuracy, Precision, Recall, macro-F1, and AUC-ROC.	Baseline
Basic	B2	Engineer features from VLE logs; apply K-means; validate with silhouette score; visualise clusters (PCA/t-SNE); interpret profiles educationally.	RQ1

Tier	Tier ID	Description	RQ Addressed
Intermediate	I1	Train profile-enriched XGBoost model (same architecture as B1, cluster label as an additional feature); compare F1 and AUC-ROC to B1.	RQ2
Intermediate	I2	Apply SHAP to both models and produce comparative interpretability analysis.	RQ3
Advanced	A1	Conduct cross-module profile transferability study.	Generalisation
Advanced	A2	Perform multi-class engagement prediction and analyse class-specific explanations.	Robustness

A Data Preparation and Problem Formulation

Before model training, the raw OULAD tables are transformed once into a single analytical dataset used by all tiers. As summarised in Table 3, the workflow includes six steps: table integration, column-specific missing-value treatment, target construction, no-VLE learner handling, primary module selection, and feature engineering with categorical encoding.

Table 4: Data Preparation and Problem Formulation workflow.

Step	Focus	Key implementation decision
1	Multi-table joining and analysis unit	Join seven OULAD tables and preserve student–module–presentation as the row-level analytical unit.
2	Missing-value handling	Apply per-column treatment to distinguish structural missingness from random gaps and add indicator flags where needed.
3	Target construction	Encode <code>Withdrawn</code> as positive class (1) and all other outcomes as class 0 for early-risk binary prediction.
4	No-VLE learner treatment	Retain no-VLE learners with zeroed VLE features as a meaningful high-risk subgroup rather than removing them.
5	Module selection strategy	Select FFF for B1–I2 based on VLE coverage, withdrawal comparability, and clickstream richness versus BBB.
6	Feature engineering and encoding	Build VLE, assessment, and demographic features; use temporal split (2013 train/2014 test) and class-imbalance weighting.

Four decisions require additional justification.

Target definition. The binary outcome label is defined as `Withdrawn` = 1 versus all remaining outcomes = 0, which aligns the learning objective with early disengagement detection under a class imbalance of approximately 32%.

No-VLE learner treatment. Students with no recorded VLE interactions (approximately 16%) are retained in the dataset and represented with zeroed VLE features; this decision preserves a substantively distinct and disproportionately high-risk subgroup that would otherwise be excluded from analysis.

Module selection. As shown in Figure 1, FFF (STEM) is selected as the primary module for B1–I2 because it has higher VLE coverage than BBB (92.2% vs 84.5%), richer clickstream activity (median 2,116 vs 454 clicks per active learner), and similar withdrawal prevalence (~30–31%). Under the required temporal split (2013 train, 2014 test), only BBB, DDD, and FFF provide complete B/J presentation coverage (AAA has no B presentation, CCC has no 2013 presentation, and EEE/GGG miss 2013B). FFF is chosen over DDD because its withdrawal rate (31.0%) is closer to BBB’s (30.2%) than DDD’s (35.9%), reducing class-balance confounding in the STEM-versus-Social-Science comparison.

Temporal split and class imbalance. Model evaluation uses a cohort-based temporal split (2013 train; 2014 test) instead of random partitioning to assess cross-cohort generalisation. Training imbalance is handled by `scale_pos_weight`, set to the negative-to-positive ratio (~2.3:1 for FFF), without resampling.

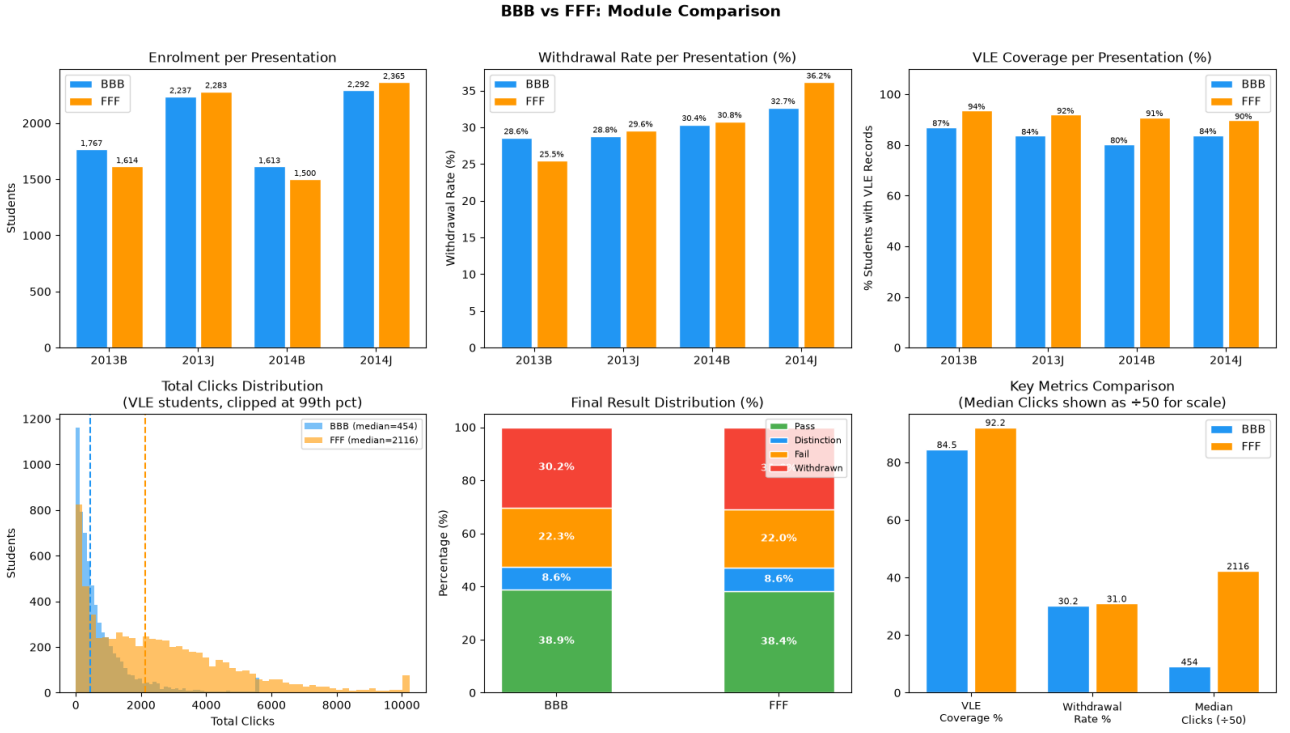


Figure 1. BBB vs. FFF: six-panel module comparison.

B B1 — Raw-Feature XGBoost Baseline

Tier B1 builds the reference model used by later tiers in three steps. First, a 12-feature matrix (VLE, assessment, demographic) is used to train XGBoost on 2013 and test on 2014. Second, `imd_band` is removed as a fairness control, yielding the 11-feature baseline used in downstream comparisons. Third, five behavioural descriptors are added to form B1+ (16 features), testing whether manual feature enrichment alone can narrow the gap without clustering.

Why XGBoost. XGBoost is used because it performs well on educational tabular data, supports interpretable feature-importance analysis consistent with SHAP in I2 (Lundberg et al. 2020), handles class imbalance through `scale_pos_weight`, and is computationally efficient on $\sim 7,700$ samples.

Why exclude `imd_band`. The `imd_band` feature captures UK postcode-level socioeconomic deprivation. Including it may inflate predictive performance by exploiting a socioeconomic proxy that reflects systemic inequality rather than learner agency or behaviour. This concern was explicitly raised by AlFaress et al. (2025), who identified `imd_band` as a dominant predictor in their OULAD model and flagged the associated algorithmic-bias risk. To isolate the contribution of purely behavioural and demographic signals, the primary baseline variant used in all downstream comparisons (I1, I2) excludes `imd_band`, reducing the feature space from 12 to 11 features. The full-feature (12-feature) model is retained only as an upper-bound reference.

Why B1+. To test whether richer hand-crafted behavioural features can improve on the compact baseline without any clustering information, a second baseline variant (B1+) augments the 11-feature set with five additional descriptors, bringing the total to 16 features. These additions are motivated by prior findings that activity-type diversity and registration timing are informative behavioural signals in distance learning.

C B2 — VLE Feature Engineering and K-Means Learner Profiling

Tier B2 answers RQ1 by constructing learner profiles used in later tiers. The design fixes the clustering unit and no-VLE treatment to match B1; builds a compact 7-feature representation of engagement volume, regularity, and distribution; selects K using multiple validity criteria; maps numeric cluster IDs to a fixed ranked archetype vocabulary; and evaluates richer B2+ and B2-Ext feature sets to test whether finer behavioural representation improves separation.

Clustering unit and no-VLE learners. The clustering unit is aligned with the FFF-centred pipeline so B2

remains comparable with B1 and feeds directly into I1. No-VLE learners are kept as an explicit profile rather than forced into behaviour-based clusters, because “no activity” is treated as a distinct high-risk state; merging them into regular clusters would reduce interpretability and intervention value.

The compact feature set. The compact B2 set targets clarity and stability: it captures engagement volume, regularity, and temporal distribution with a deliberately small feature space to keep archetypes interpretable. Distribution control and scaling are used to limit dominance by extreme users. This is the primary RQ1 candidate feature design.

Choosing the number of clusters. Cluster count is selected as a design choice, not preset, using multiple criteria (compactness, separation, overlap) to reduce reliance on any single validity score.

B2+: adding behavioural context. B2+ tests whether a finer representation adds useful separation by introducing activity-type and timing descriptors while controlling redundant volume signals.

B2-Ext: adding temporal-trend features. B2-Ext further tests whether temporal trend and timeliness improve separation by adding four features to B2+: `engagement_slope`, longest inactivity gap, missing formative assessment count, and average submission lead time. Compact, B2+, and B2-Ext are compared for RQ1 using silhouette and Davies–Bouldin across a range of K .

Naming the archetypes. Archetype naming translates arbitrary cluster IDs into a fixed ranked engagement vocabulary so interpretation stays consistent across tables, figures, and discussion. *No VLE Record* remains an explicit archetype for practical intervention relevance.

D I1 — Profile-Enriched XGBoost

Tier I1 answers RQ2 with a controlled augmentation: keep the predictive framework fixed and add only profile identity from B2. Pair A adds `cluster_p` to the compact baseline, and Pair B adds it to B1+, testing whether profile signal helps both generally and under richer manual features. Design controls cover encoding consistency, fixed model configuration, and formal significance testing.

Encoding `cluster_p`. `cluster_p` is the integer archetype index from the selected B2 specification. It is added as one extra column (no one-hot expansion) to test minimal profile injection without extra dimensionality. The same B2 assignment logic is applied to train and test cohorts so semantics remain consistent across paired comparisons.

Holding model configuration constant. All four models (B1, I1, B1+, I1+) use the same tuned XGBoost configuration inherited from B1. This avoids confounding profile contribution with re-tuning effects and keeps paired comparisons methodologically fair.

Testing statistical significance. Observed pair differences are tested with McNemar’s exact test on discordant predictions from the same 2014 cohort, matching I1’s paired design. A two-sided $\alpha = 0.05$ is used for both pairs.

E I2 — SHAP Interpretability

Tier I2 answers RQ3 by auditing model behaviour at feature and archetype levels using the same four variants as I1 (B1, I1, B1+, I1+). SHAP is used because `shap.TreeExplainer` provides exact, additive attributions for tree ensembles at both global and instance levels (Lundberg & Lee 2017, Lundberg et al. 2020). The workflow has three steps.

Step 1: global feature ranking. The first step ranks features by overall importance. For each model, the mean absolute SHAP value is computed across the test set: $|\bar{\phi}_j| = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}|$, where ϕ_{ij} is the SHAP value of feature j for instance i . Rankings and mean $|\text{SHAP}|$ of `cluster_p` in I1 and I1+ quantify its contribution relative to established behavioural features.

Step 2: per-archetype consistency check. The second step checks consistency of `cluster_p` within archetypes (Disengaged, Moderate, High Engaged, No VLE Record). For each archetype, distributions of `cluster_p`-specific SHAP values are compared between I1 (Pair A) and I1+ (Pair B), and their Pearson correlation $r_{\text{cluster_p}}$ is used as a stability check. A waterfall example illustrates interaction with other features in one prediction.

Step 3: archetype-level driver ranking.

The third step identifies drivers beyond `cluster_p` for each archetype. I1 SHAP values are split by archetype, and mean absolute SHAP is recalculated within each group to obtain archetype-level rankings. This

tests whether risk is driven by the same features across archetypes, or by secondary signals hidden in an overall ranking.

F A1 — Cross-Module Transferability

What is tested. Tier A1 tests whether the full FFF-trained pipeline (B2+ K-Means, StandardScaler, I1 XGBoost) generalises to a structurally different module (BBB) *without retraining*.

Evaluation metrics. Performance is evaluated with the same five metrics as B1–I1 (Accuracy, macro-Precision, macro-Recall, macro-F1, AUC-ROC), recomputed on BBB for direct comparability.

Distinguishing miscalibration from signal loss. Any metric drop on BBB can arise from miscalibration or genuine signal loss. To separate these cases, post-hoc isotonic calibration and a decision-threshold scan are fitted on a small BBB calibration split.

Checking whether the profiles still make sense. Independently of prediction accuracy, profile validity on BBB is checked by comparing BBB and FFF VLE engagement distributions. Because frozen B2+ centroids are fitted on FFF, a large distribution shift would suggest FFF-specific rather than module-general profile semantics.

G A2 — Multi-class problem formulation

Task setup. Tier A2 is a robustness check that tests whether `cluster_p` still helps when the task is harder: predicting all four `final_result` classes (0 Distinction, 1 Fail, 2 Pass, 3 Withdrawn) instead of a binary label. No categories are collapsed, and the 2013/2014 temporal split is kept identical to B1–I1 for comparability.

Model configuration. Two models are trained: B1+ multi-class baseline (16 features) and I1+ multi-class model (17 features = B1+ + `cluster_p`), both with `XGBClassifier` (`objective='multi:softprob'`) and the tuned B1 hyperparameters so performance differences are attributable to `cluster_p` (Chen & Guestrin 2016).

Handling class imbalance. Because Distinction is under-represented in FFF training, balanced per-sample weights (`compute_class_weight`) are passed as `sample_weight` to prevent majority-class collapse.

Explainability check. SHAP TreeExplainer is applied to both models, with per-class global importance from class-wise mean absolute SHAP values (Lundberg & Lee 2017, Lundberg et al. 2020). The `cluster_p` contribution is additionally summarised in a `cluster` $\times$ `class` mean-SHAP heat-map to test whether archetype effects are class-consistent or class-specific.

H Summary of Solution

The six tiers build cumulative evidence for the three research questions rather than optimising a single headline metric. B1/B1+ set predictive baselines without clustering; B2 compares compact, B2+, and B2-Ext feature designs to answer RQ1; I1 tests incremental value of `cluster_p` for RQ2 with paired significance testing; I2 evaluates cluster-linked explanations for RQ3; and A1 plus A2 stress-test generalisation and robustness under cross-module transfer and multi-class prediction.

IV RESULTS AND EVALUATION

This section reports the outcomes of the experiments and evaluates how far they answer the research questions. The results show that the proposed pipeline does not simply produce a higher score; rather, it indicates that behavioural clustering can generate meaningful learner profiles, that these profiles provide small but measurable predictive value beyond strong raw-feature baselines, and that the resulting model can offer interpretable evidence for at-risk decisions. The evaluation therefore focuses on what the experiments actually revealed: whether the profiles are meaningful, whether they improve prediction, and whether the explanations are useful in practice. At the same time, the two checks in the Advanced tier qualify this picture: the transfer experiment shows that the approach is not yet fully generalisable across modules, and the robustness experiment shows that `cluster_p`'s predictive and explanatory value does not survive the harder multi-class task; both should be treated as limitations rather than failures of the method.

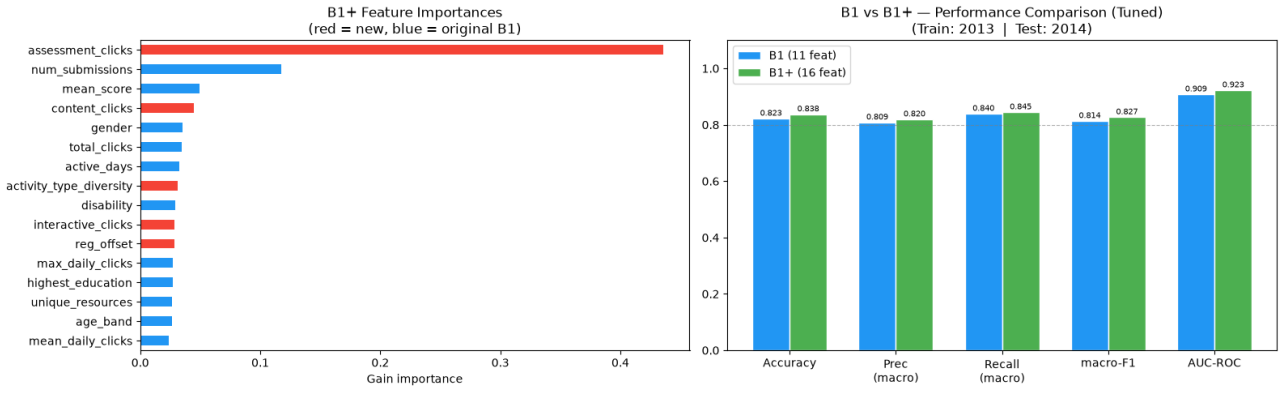


Figure 2. B1 vs. B1+: baseline performance comparison.

A B1 — Raw-Feature Baseline Performance

B1 baseline performance. The raw-feature XGBoost baseline achieved macro-F1 = 0.8140 and AUC-ROC = 0.9085. In practical terms, this means the model is already good at telling withdrawn students apart from those who stay on the course, and it rarely misses a genuinely at-risk student. Before asking whether learner profiles add anything useful, it is first necessary to confirm that the baseline they are compared against is not artificially weak. Because B1 performs well on its own, any later improvement from adding cluster information can be interpreted as a directional improvement rather than an artefact of a poor starting point. With macro-F1 above 0.81 and AUC-ROC above 0.90, B1 satisfies this precondition, establishing a strong, credible raw-feature baseline that grounds every later comparison in the pipeline.

Comparison with related work. Read against the state-of-the-art surveyed in Section C, B1 achieves accuracy = 0.8225, macro-F1 = 0.8140, and AUC-ROC = 0.9085, which is broadly comparable to prior OULAD-based models despite using a smaller feature set and a harder protocol. On accuracy, Torkhani & Rezgui (2025) report 82.73% (Random Forest) and 83.41% (LSTM) on a single AAA module (748 students), close to B1’s 82.25%; AlFaress et al. (2025) report 91.02% accuracy and F1 = 91.18% but with a 44-feature set (about four times B1’s 11 features); and Mansouri et al. (2026) report 90% accuracy using a SMOTE-GA-GRU pipeline with limited interpretability. B1 is therefore within roughly 1–9 percentage points of these benchmarks while remaining compact and interpretable. Two caveats are important: first, with an imbalanced outcome (about 30% Withdrawn), accuracy alone can be optimistic, so macro-F1 is reported as the stricter metric; second, B1 uses a harder cross-cohort split (2013 train, 2014 test), whereas many studies use random within-cohort splits. Under this stricter setting, and without the interpretability cost noted for deep models by Mansouri et al. (2026) and Waheed et al. (2022), B1 remains a credible and directly comparable baseline.

B1+ extended baseline. The extended baseline B1+ pushes performance further, to macro-F1 = 0.8273 and AUC-ROC = 0.9231 — an improvement of +1.33 percentage points and +1.46 percentage points respectively over B1. This gain comes from adding a handful of richer behavioural descriptors, such as how varied a student’s activity types are and when they registered for the course, rather than from any clustering information. The practical implication is that B1+ sets a tougher benchmark: whatever advantage Tier II later claims for cluster-derived profiles must be measured against this stronger, not the weaker, baseline. Figure 2 compares the two baselines side by side and shows that B1+ outperforms B1 across every reported metric.

Limitations. Nevertheless, B1 has two limitations that motivate the rest of the pipeline.

1. *No concept of learner subgroups.* The model treats a mostly-engaged student who missed one assignment the same way as a student who never logged in at all, even though these two cases call for very different kinds of support.
2. *Cross-cohort evaluation.* Because training and testing use different academic years, the reported performance reflects generalisation across cohorts rather than within a single one; this is a more realistic test, but it also means that shifts from one year to the next cannot be fully ruled out.

Together, these two points explain why Tier II introduces cluster-derived learner profiles: to give the model a sense of learner subgroup that raw features alone cannot supply.

B B2 — Cluster Profiling Outcome

B2’s results directly answer RQ1: a compact 7-feature set, describing only how much a student interacts, how many days they are active, and how that activity is spread over time, outperforms two richer alternatives tested. Clustering it with K-means at $K = 3$ produces three behavioural archetypes — Disengaged, Moderate, and High Engaged — and, together with a separately retained No VLE Record group for learners with no clickstream activity, gives four operational profiles that are reused throughout every downstream tier. The remainder of this subsection builds up this answer in three steps: first, which feature set separates learners best; second, whether that separation is educationally meaningful rather than a statistical artefact; and third, why $K = 3$ is chosen over the statistically optimal $K = 2$. It closes with the limitations that bound how far the answer should be trusted.

Step 1: which feature set separates learners best. Three feature sets are compared under the same K-means setup and $K = 2, 3, 4$ sweep (Table 4): compact B2 (7 features; one row per student), B2+ (12 features), and B2-Ext (16 features; both one row per student–presentation pair). Because observation units differ, the samples are not identical; this is retained as an explicit limitation. Even so, the compact set is best at every tested K , with the highest silhouette and lowest Davies–Bouldin. At $K = 2$, silhouette is 0.4329 (B2) versus 0.3684 (B2+) and 0.3673 (B2-Ext), while Davies–Bouldin is 0.8466 versus 1.0616 and 1.1341; the same ranking holds at $K = 3$, the operational choice. Adding activity-type, regularity, assessment-participation, and registration-timing features therefore blurs rather than sharpens the clusters, and the compact set remains the strongest separator across all tested K .

Table 5: Cross-model K-selection summary: silhouette score and Davies–Bouldin (DB) Index for the compact B2 (7-feature), enriched B2+ (12-feature), and further-enriched B2-Ext (16-feature) sets at $K = 2, 3, 4$.

Model	K	Silhouette $\uparrow$	DB $\downarrow$
B2 (7 feat)	2	0.4329	0.8466
	3	0.3443	1.1472
	4	0.3500	1.0119
B2+ (12 feat)	2	0.3684	1.0616
	3	0.2697	1.4713
	4	0.2619	1.3875
B2-Ext (16 feat)	2	0.3673	1.1341
	3	0.3098	1.2821
	4	0.2387	1.5036

Step 2: is the separation educationally meaningful? Strong silhouette and Davies–Bouldin values alone do not guarantee educational meaning, so this step validates clusters against withdrawal, which is not used in clustering. Withdrawal rates differ sharply across High Engaged, Moderate, and Disengaged groups, and PCA shows these groups occupying distinct regions rather than overlapping heavily. In the FFF 2013 training cohort, the No VLE Record group is also non-trivial in size and clearly distinct from the three K-means clusters; its withdrawal rate lies far outside the range of other archetypes, supporting no VLE activity as a real risk category rather than noise. Since withdrawal rate and enrolment share are external to the clustering objective, their agreement with the geometric separation provides independent evidence that the profiles capture non-random behavioural structure.

Step 3: why $K = 3$ instead of the statistically optimal $K = 2$. The cluster validation plots show that $K = 2$, not $K = 3$, is the silhouette- and Davies–Bouldin-optimal partition for both the compact and enriched feature sets, corresponding to a coarse Engaged/Disengaged split (Figure 3). A binary partition, however, is educationally crude: it cannot tell apart a learner who engages steadily at a moderate level from one who engages intensely only around assessment deadlines. $K = 3$ is therefore chosen over the statistically optimal $K = 2$, trading a modest drop in silhouette score and Davies–Bouldin Index for a partition that is far more useful for identifying at-risk learners.

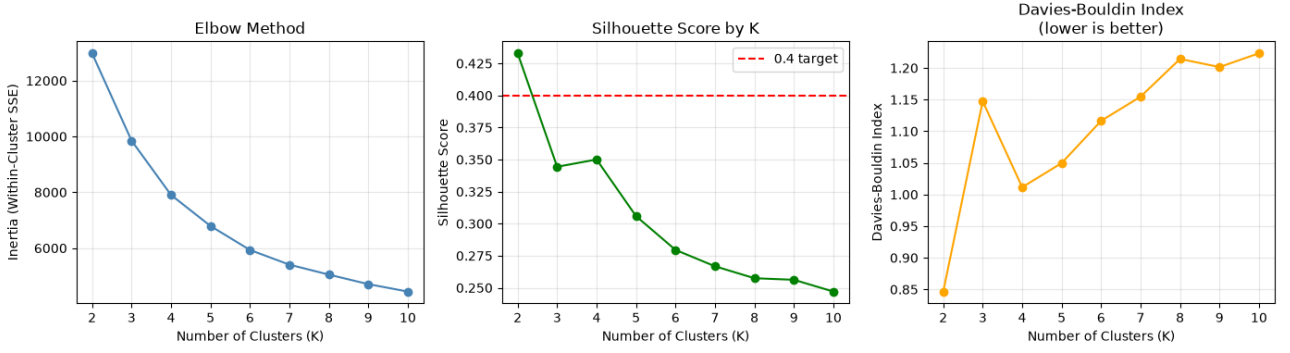


Figure 3. B2 cluster-count selection: elbow (inertia), silhouette score, and Davies–Bouldin Index across $K = 2$ – 10 . $K = 2$ is statistically optimal on all three criteria, but $K = 3$ is selected for its educational interpretability.

Limitations. These results support the RQ1 answer above, but three limitations set bounds on how far it should be trusted.

1. *Single-module fitting.* The cluster solution comes from the FFF module training set only; the A1 transfer experiment later shows that the same centroids do not map cleanly onto BBB learners, suggesting the profiles capture module-specific engagement norms rather than universal archetypes.
2. *Unequal sample units in Table 4.* The compact set’s advantage is measured against B2+ and B2-Ext at a different observation unit and therefore a different sample size, one row per student against one row per student–presentation pair; part of the compact set’s higher silhouette score may reflect this smaller, less repetitive sample rather than the informativeness of its seven features alone, so the ranking in Table 4 should be read as suggestive rather than as a fully controlled comparison.
3. *The downstream label is not the best-separated one.* The `cluster_p` feature used throughout the downstream I1, I2, A1, and A2 tiers comes from the statistically weaker B2+ (12-feature) partition rather than the better-separated compact B2 (7-feature) partition identified above, even though the 7-feature set is the answer to RQ1. As explained in the Solution section, this choice is deliberate rather than an oversight: the compact set’s per-student granularity is incompatible with the per-student–presentation unit used by every downstream tier, and B2+ shares that unit and several behavioural signals with the B1+ baseline, keeping the I1+ comparison clean. Choosing B2+ therefore favours internal consistency of the downstream comparisons over maximising clustering quality, and the profile labels used for downstream prediction and explanation are consequently not the most cleanly separated archetypes the data can produce.

C I1 — Profile-Enriched Binary Prediction

RQ2 asks whether adding cluster-derived profile membership improves early at-risk prediction performance compared with strong raw-feature baselines. The answer is directionally yes, but the size of the improvement depends on how strong the baseline already is: adding `cluster_p` as a single extra feature raises both macro-F1 and AUC-ROC in every comparison tested, yet the gain shrinks once the baseline itself already contains richer hand-crafted behavioural descriptors. Confirming this answer needs two matched comparisons rather than one, because a single model pair cannot separate a profile-linked effect from an effect that only appears alongside one particular baseline. Pair A therefore adds `cluster_p` to the compact 11-feature B1 baseline, and Pair B adds the same label to the stronger 16-feature B1+ baseline; both pairs share the same XGBoost configuration, temporal split, and evaluation protocol, so any difference between the two pairs can be attributed to the baseline’s starting strength rather than to a change in method. The rest of this subsection reports each pair in turn, draws the two together to answer RQ2, and closes with the limits on how far that answer should be trusted.

Pair A: adding `cluster_p` to the compact B1 baseline. Adding `cluster_p` to the 11-feature B1 baseline lifts macro-F1 from 0.8140 to 0.8210 (+0.0070) and AUC-ROC from 0.9085 to 0.9161 (+0.0076); both metrics move in the same direction, so the cluster label is not simply trading recall for precision. Figure 4 shows the underlying evidence: the left panel ranks feature importance for the 12-feature I1 model and places `cluster_p`

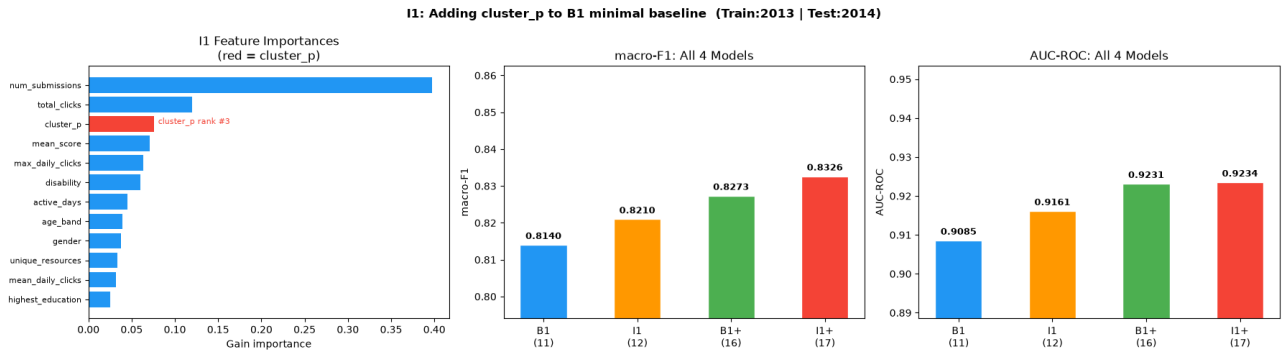


Figure 4. Pair A — adding `cluster_p` to the minimal B1 baseline: I1 feature importances (`cluster_p` ranked 3rd of 12) and macro-F1/AUC-ROC comparison across all four models.

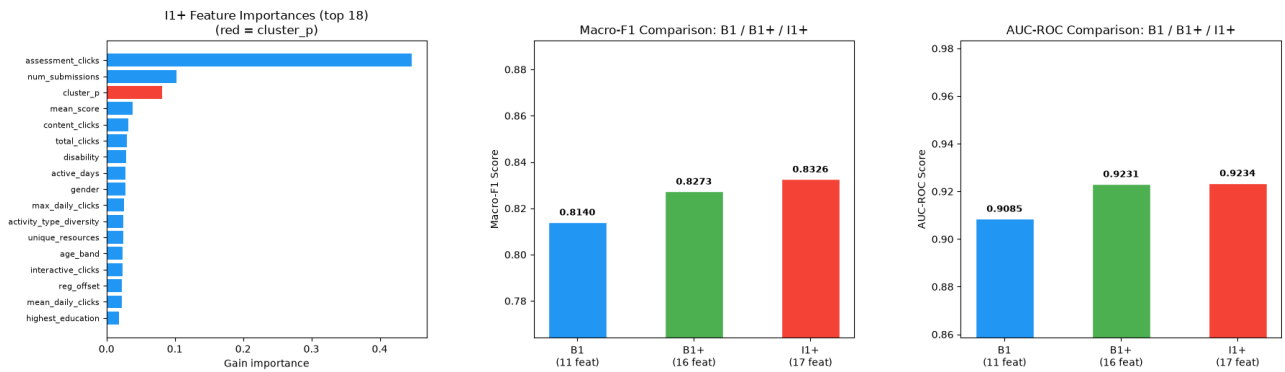


Figure 5. Pair B — adding `cluster_p` to the enriched B1+ baseline: I1+ feature importances (`cluster_p` ranked 3rd of 17) and macro-F1/AUC-ROC comparison across B1, B1+, and I1+.

third overall, immediately behind `num_submissions` and `total_clicks` and ahead of every demographic and secondary VLE feature; the middle and right panels place all four models side by side on macro-F1 and AUC-ROC, confirming that I1 sits strictly between B1 and B1+ on both axes. A McNemar exact test on the paired 2014 predictions gives $p = 0.0903$, which falls just short of the conventional 0.05 threshold; the improvement is therefore consistent and visible on both metrics, but not yet strong enough to rule out chance at standard significance levels.

Pair B: adding `cluster_p` to the enriched B1+ baseline. The same `cluster_p` feature is added to the stronger 16-feature B1+ baseline. Macro-F1 rises from 0.8273 to 0.8326 (+0.0053) and AUC-ROC rises from 0.9231 to 0.9234 (+0.0003). Figure 5 shows that `cluster_p` again ranks third, now among 17 features in the I1+ model, behind only `assessment_clicks` and `num_submissions`, and that I1+ edges ahead of B1+ on both the macro-F1 and AUC-ROC comparison panels. The corresponding McNemar test gives $p = 0.1425$, further from significance than Pair A’s result. Read together, the two pairs describe a diminishing-returns pattern: the absolute and statistical strength of the cluster-label effect is larger against the compact baseline than against the already-enriched one, suggesting that part of the information carried by `cluster_p` overlaps with the hand-crafted features already present in B1+.

Synthesis. Altogether, Pair A and Pair B answer RQ2 in the affirmative: cluster-derived profile membership improves prediction over both a minimal and an enriched raw-feature baseline, and it does so consistently on macro-F1 and AUC-ROC rather than trading one for the other. `cluster_p`’s stable rank-3 position across two feature sets of different size and composition is itself informative: it means the behavioural summary captured by clustering is not an artefact of one particular feature combination, but a signal the tree ensemble finds useful whichever hand-crafted features happen to sit alongside it.

Limitations. Two limitations qualify this answer.

1. *No statistically significant McNemar test.* Neither test reaches the conventional 0.05 significance threshold, so the improvement, while consistent in direction and magnitude across both pairs, should be read as suggestive rather than conclusive; a larger test cohort or a repeated cross-validation protocol would be

needed before treating the gain as statistically established.

2. *Unordered integer encoding.* The cluster label is encoded as an unordered integer, so the tree ensemble must rediscover any ordinal structure among the profiles from scratch rather than being told directly that engagement rises from Disengaged through Moderate to High Engaged; an ordinal or probability-weighted encoding of `cluster_p` is left for future work and might recover a larger share of the profile’s predictive value.

D I2 — Explainability Analysis

RQ3 asks two linked questions, so the results below are organised to answer each in turn: first, whether SHAP-based feature importance shifts systematically across learner archetypes and across the Pair A/Pair B baselines; second, which specific behavioural features — beyond `cluster_p` itself — drive the withdrawal prediction within each archetype. Three sub-stages are reported in turn (global ranking, `cluster_p` deep dive, per-archetype ranking), followed by a synthesis that answers both halves of RQ3 directly.

Step 1: global feature ranking shifts across baselines. For Pair A, SHAP mean-absolute ranking places `cluster_p` second of twelve features (Mean $|\phi| = 0.6535$, 13.6% of total SHAP mass), immediately behind `num_submissions` and ahead of `mean_score` — suggesting that the cluster label captures a distinct slice of predictive variance rather than duplicating existing behavioural features. For Pair B, the picture is different: `cluster_p` drops to tenth of seventeen features (Mean $|\phi| = 0.2110$, only 2.9% of total SHAP mass), even though the same feature ranked third by XGBoost’s built-in gain importance in Tier I1 (Section C). Once richer hand-crafted descriptors such as `assessment_clicks` and `content_clicks` are already present, `cluster_p` has little additional log-odds mass left to contribute on a per-instance basis. This gain-versus-SHAP disagreement is itself informative: gain measures how often a feature is used to split trees, whereas mean $|\phi|$ measures the typical magnitude of its per-instance contribution, so a feature can be selected reasonably often (gain rank 3) while still contributing a small, uniform push to any single prediction (SHAP rank 10). Read alongside the McNemar results in Tier I1, this reinforces the diminishing-returns picture already established there: `cluster_p` adds real but modest value once B1+’s five additional descriptors are already present.

Step 2: does `cluster_p` behave consistently by archetype? Figure 6 compares `cluster_p` SHAP values by archetype for I1 (Pair A) and I1+ (Pair B). In I1, No VLE Record has a strongly positive mean SHAP (+2.5249), showing that complete VLE absence pushes predictions toward Withdrawn, while VLE-active archetypes are negative (Moderate −0.6874, High Engaged −0.4982, Disengaged −0.1221), i.e., mildly protective conditional on the other eleven features. This ordering does not mirror raw withdrawal rates: Disengaged has the highest withdrawal among active archetypes (63.7% vs 7.3% Moderate and 1.3% High Engaged) but the least negative `cluster_p` SHAP, implying much of its risk is already captured by other features (notably `num_submissions` and `mean_score`). Cross-pair agreement is strong ($r = 0.797$), indicating stable directionality across feature sets despite different SHAP magnitudes.

Figure 7 shows the same mechanism at case level: for the highest-risk Withdrawn student in test 2014 ($P(\text{Withdrawn}) = 0.995$, No VLE Record), `cluster_p` contributes strongly but jointly with other aligned features, confirming a multi-factor explanation rather than a single-feature effect.

Step 3: archetype-level driver ranking. Table 5 and Figure 8 address the second half of RQ3 by ranking the other eleven drivers within each archetype. `num_submissions` is top-ranked in all archetypes, making it the only universal driver. Beyond this shared anchor, rankings diverge: `cluster_p` is top-3 for Moderate, High Engaged, and No VLE Record, but not for Disengaged, where `mean_score` and `unique_resources` dominate, consistent with Step 2.

Table 6: Top-3 SHAP drivers per learner archetype (I1 model, 12 feat, test set 2014).

Archetype	WD Rate	Top-3 SHAP drivers
Moderate	7.3%	<code>num_submissions</code> , <code>cluster_p</code> , <code>max_daily_clicks</code>
Disengaged	63.7%	<code>num_submissions</code> , <code>mean_score</code> , <code>unique_resources</code>
High Engaged	1.3%	<code>num_submissions</code> , <code>active_days</code> , <code>cluster_p</code>
No VLE Record	95.0%	<code>cluster_p</code> , <code>num_submissions</code> , <code>unique_resources</code>

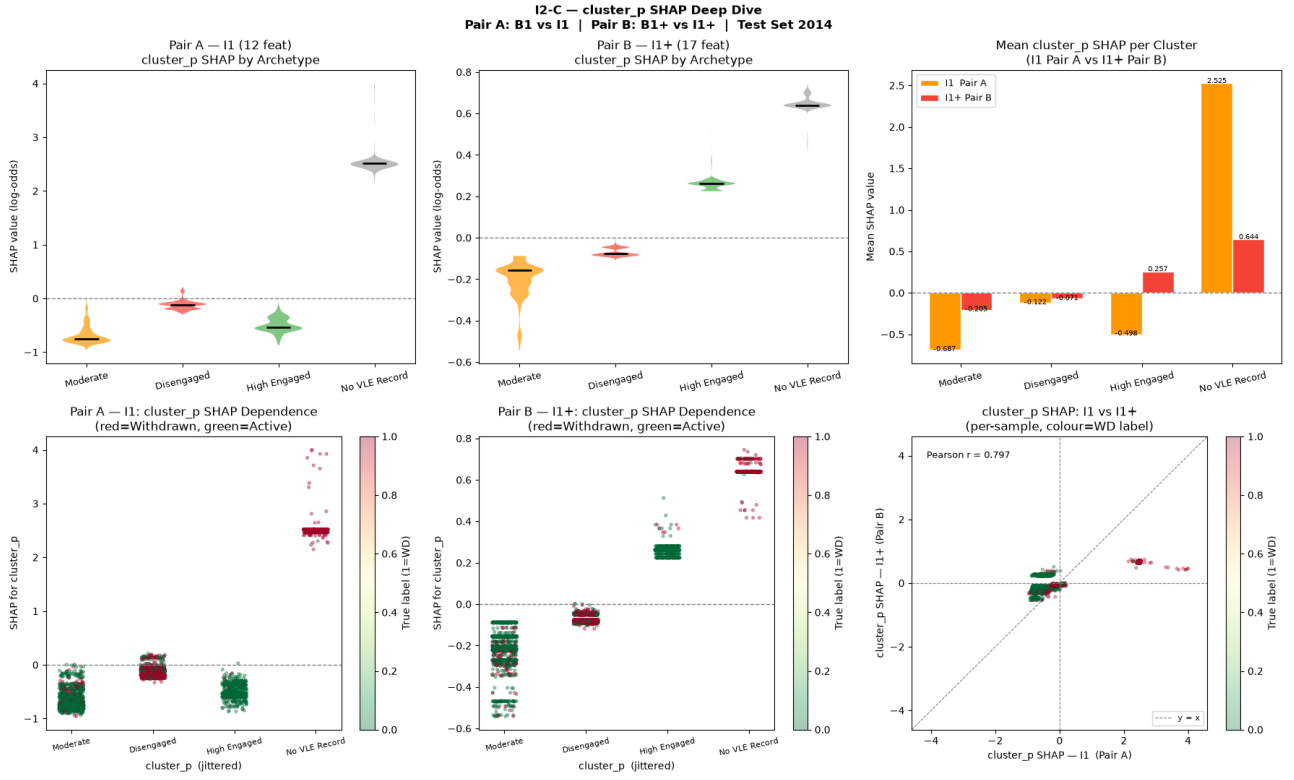


Figure 6. `cluster_p` SHAP deep dive: per-archetype violin distributions (I1, I1+), mean SHAP per archetype, per-archetype dependence scatter, and per-sample I1-vs-I1+ consistency ($r = 0.797$). Test set 2014.

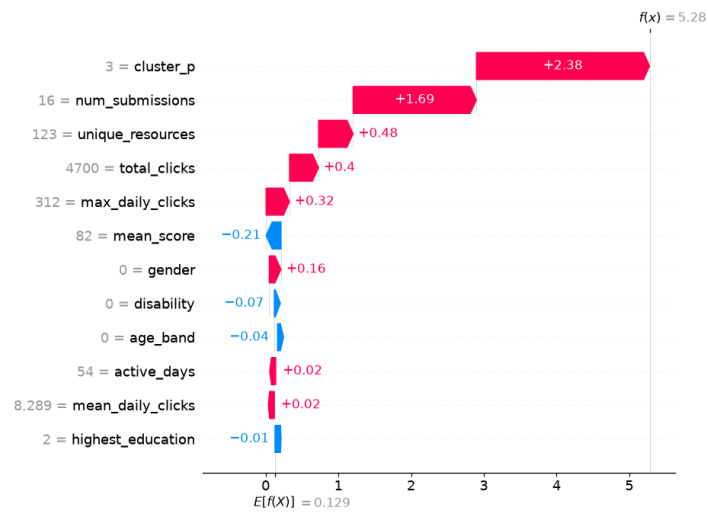


Figure 7. SHAP waterfall decomposition for the highest-risk Withdrawn student in the I1 model (test set 2014, $P(\text{Withdrawn}) = 0.995$, `cluster_p` = No VLE Record).

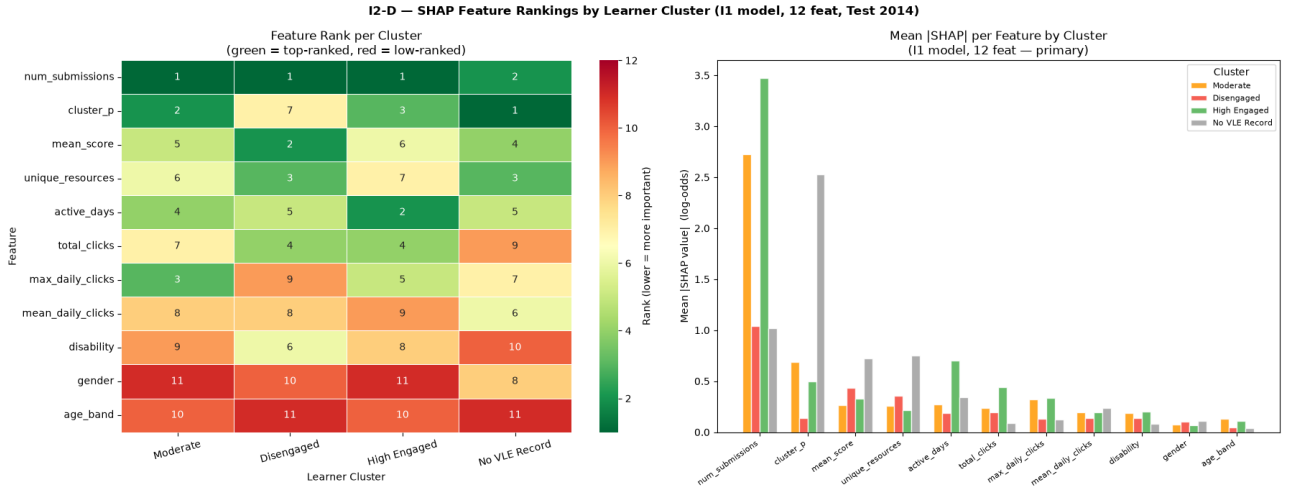


Figure 8. SHAP feature ranking by learner archetype (I1 model, 12 feat, test set 2014): rank heat-map (left) and mean $|\phi|$ grouped bars (right).

Figure 8 visualises this via a rank heat-map and grouped mean- $|\phi|$ bars. Kendall’s τ between archetype-specific rankings over the eleven shared behavioural features shows strong agreement in five of six pairs ($\tau = 0.600$ – 0.818 , $p < 0.01$), with only moderate agreement for High Engaged vs No VLE Record ($\tau = 0.491$, $p = 0.041$). Thus, most archetypes share a common behavioural hierarchy, while the two extreme archetypes are where reordering is most likely.

Synthesis. In conclusion, these three sub-stages answer RQ3 in full. On the first half of the question, SHAP rankings do differ systematically across learner profiles: `cluster_p`’s own marginal contribution is strongly positive for No VLE Record and mildly negative for the three VLE-active archetypes, this ordering is stable across Pair A and Pair B ($r = 0.797$), and the feature’s overall importance rank itself shifts from 2nd (Pair A) to 10th (Pair B) once richer hand-crafted features are already present. On the second half, the profile-specific behavioural drivers are not identical across archetypes: `num_submissions` is the one universal driver, but `cluster_p` itself re-enters as a top-3 driver for three of the four archetypes, while Disengaged learners are instead driven primarily by `mean_score` and `unique_resources` — consistent with the Kendall τ evidence that High Engaged and No VLE Record are the pair whose behavioural hierarchy diverges most.

Limitations. Three limitations qualify these findings.

1. *Association, not causation.* SHAP measures statistical association rather than causal mechanisms. A high SHAP value for `cluster_p` in the No VLE Record group tells us that the model relies heavily on that profile to predict withdrawal, but it does not tell us why those students never engaged with the VLE in the first place, which is the educationally actionable question.
2. *Gain-versus-SHAP disagreement.* The disagreement between gain-based and SHAP-based importance observed for Pair B means that any single importance measure should be read as one lens on the model rather than a complete description of `cluster_p`’s role.
3. *Single cohort, single module.* The current explanations are produced at the model level on the 2014 test cohort; whether the same SHAP pattern would hold for a future cohort or a different module is not yet verified, as the A1 transfer experiment already shows that profile structure does not transfer cleanly across modules.

E A1 — Cross-Module Transferability

A1 tests whether the FFF-trained pipeline can be applied directly to BBB without retraining. This is not a direct test of RQ1 or RQ2, both of which are scoped to within-module behaviour; rather, A1 probes a boundary condition on the B2 and I1 findings by asking whether the VLE-derived profiles and the profile-enriched classifier generalise to a module on which they were never fitted.

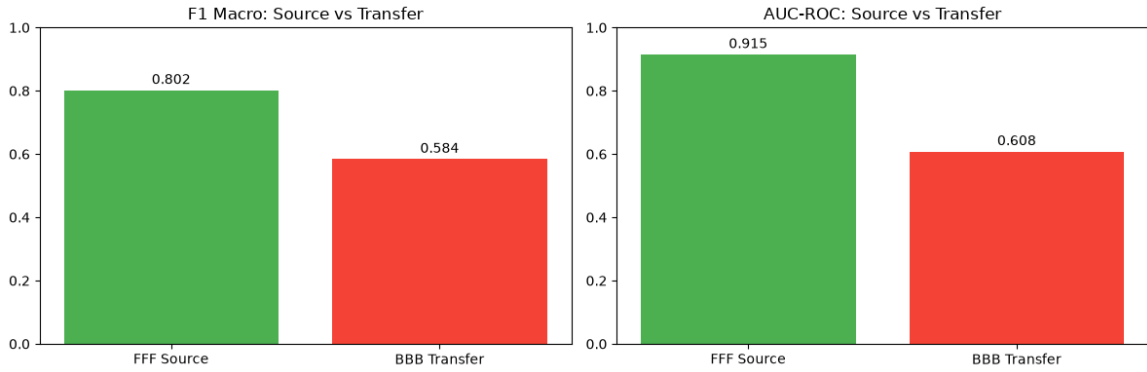


Figure 9. Zero-shot transfer of the FFF-trained I1 pipeline to BBB: macro-F1 (left) and AUC-ROC (right), FFF source vs. BBB transfer.

Result: zero-shot transfer fails. The FFF-trained I1 pipeline achieves macro-F1 = 0.802 and AUC-ROC = 0.915 in-domain, but applied directly to BBB without retraining it produces Accuracy = 0.6173, macro-Precision = 0.5864, macro-Recall = 0.5996, macro-F1 = 0.5842, and AUC-ROC = 0.6081: a drop of 21.8 percentage points on macro-F1 (0.802 $\rightarrow$ 0.584) and 30.7 percentage points on AUC-ROC (0.915 $\rightarrow$ 0.608).

Why the AUC-ROC drop is larger. Figure 9 explains why the AUC-ROC drop is proportionally the larger of the two: an AUC-ROC of 0.608 is only marginally better than the 0.5 random-ranking baseline, which indicates that the model’s probability ranking of at-risk students collapses much more severely under transfer than its hard-label accuracy does. This asymmetry is consistent with a calibration failure rather than a pure loss of discriminative signal — the model still separates classes somewhat better than chance, but its probability scale no longer reflects BBB’s actual risk ordering, which is exactly the miscalibration that the post-hoc isotonic calibration and threshold scan (described above) are designed to partially correct. Even after that calibration and threshold-optimisation step, however, the corrected macro-F1 remains far below the FFF in-domain level, showing that recalibration can only compensate for the probability shift, not for the underlying loss of cluster validity.

Root cause: domain shift. Figure 1 explains the deeper cause: domain shift. BBB students have a substantially lower median VLE click count than FFF students (454 vs. 2116), so the FFF cluster centroids — fitted on a much more active population — no longer partition the BBB engagement distribution in a behaviourally meaningful way; a BBB student who is only moderately engaged by BBB norms can be assigned to a low-engagement archetype calibrated on FFF’s higher activity baseline.

Summary. The profiles discovered in B2 are partly module-specific rather than universal archetypes, and cross-module deployment requires at minimum a re-fit of the clustering and scaler on the target module’s training data. The A1 experiment therefore serves as an important qualification of the B2 and I1 results: strong in-domain performance does not automatically imply transferability, and any practical deployment of this pipeline to a new module should budget for at least a lightweight re-calibration stage rather than assuming zero-shot reuse is sufficient.

F A2 — Multi-class Outcome Prediction

A2 is a robustness check: it asks whether `cluster_p` still adds value once the task is made harder, by predicting all four OULAD outcomes (Distinction, Fail, Pass, Withdrawn) instead of the binary at-risk label used in I1.

Overview: the check fails on all three fronts it was designed to test. First, on *performance*: the B1+ baseline achieves macro-F1 = 0.6911, and adding `cluster_p` (I1+) produces macro-F1 = 0.6885 — a small *decrease* of 0.0026, rather than the improvement seen in the binary I1 setting. Second, on *importance*: across the 17 I1+ features, `cluster_p` ranks only 12th by mean |SHAP|, far below its binary-setting standing. Third, on *direction*: the archetype-level directional coherence documented for binary withdrawal in I2 does not hold once the model must separate all four outcomes, and holds uniformly only for the No VLE Record archetype.

Importance. Figure 10 explains the drop in importance. Once `mean_score` and `num_submissions` are available to separate four outcomes rather than one, `cluster_p`’s contribution shrinks sharply: it ranks 12th of 17 features by mean |SHAP|. It is far behind `mean_score` (0.778), `num_submissions` (0.733), and

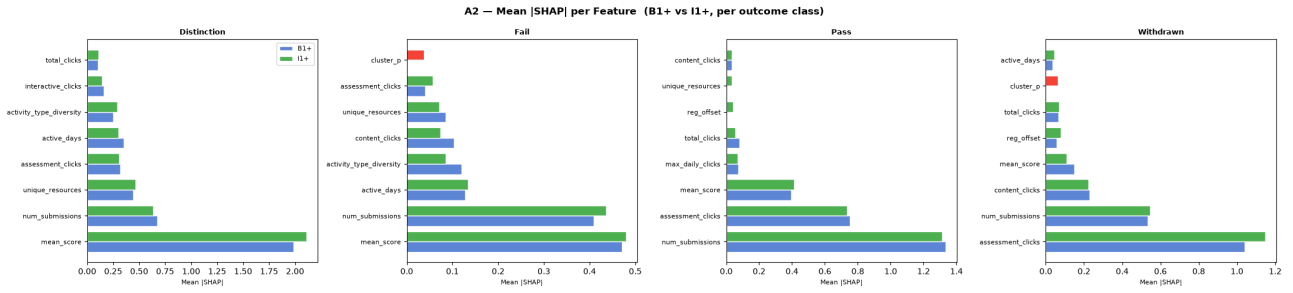


Figure 10. Mean |SHAP| per feature by outcome class, B1+ vs. I1+. `cluster_p` (red bar) ranks 12th of 17 features overall and contributes zero mean |SHAP| to Distinction and Pass.

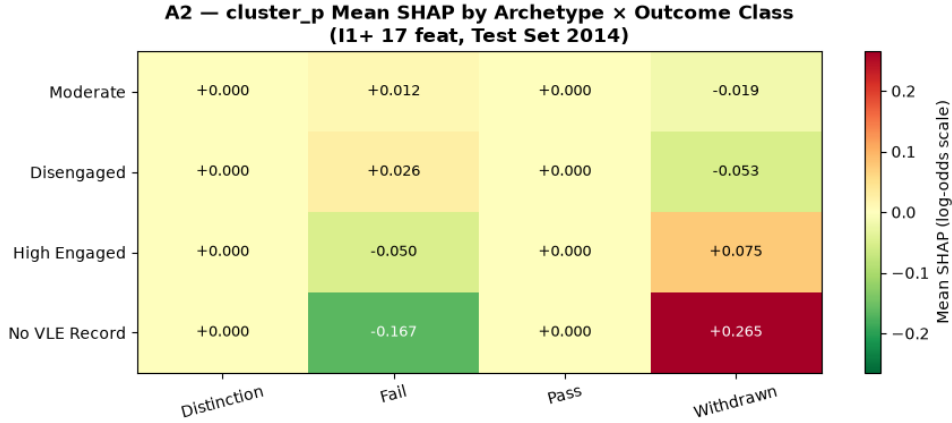


Figure 11. `cluster_p` mean SHAP by archetype $\times$ outcome class (I1+, test 2014). Only No VLE Record shows the expected strong positive push toward Withdrawn.

`assessment_clicks` (0.562), and contributes essentially zero mean |SHAP| to Distinction and Pass, which are driven mostly by assessment features that a binary withdrawal model did not need to separate. Its residual contribution is confined to Fail (mean |SHAP| = 0.038) and Withdrawn (mean |SHAP| = 0.063), consistent with the profile structure identified in B2, but too small in absolute terms to move macro-F1.

Direction. Figure 11 explains the breakdown in direction, by cross-tabulating `cluster_p`'s mean SHAP value against outcome class for each archetype. Only the No VLE Record archetype (95% empirical withdrawal rate) behaves as the design predicted, pushing strongly toward Withdrawn (mean SHAP = +0.265) and away from Fail (-0.167). The other three archetypes do not follow a monotonic risk ordering: Moderate (7.3% withdrawal rate) and Disengaged (63.7% withdrawal rate) both show a small *negative* mean SHAP toward Withdrawn (-0.019 and -0.053 respectively), while High Engaged (1.3% withdrawal rate, the safest archetype) shows a small *positive* mean SHAP toward Withdrawn (+0.075). Once the other 16 features already capture most of the raw engagement signal, `cluster_p`'s marginal, model-relative contribution to the Withdrawn class no longer tracks the archetypes' raw empirical risk ordering except at the No VLE Record extreme.

Summary. The honest reading of A2 is that `cluster_p`'s value and its behavioural interpretation are both binary-task-specific rather than general properties of the profiles: the gain reported for RQ2 and the directional story reported for RQ3 should not be assumed to hold in any setting more complex than the one in which they were demonstrated. The one exception is the No VLE Record archetype, whose behaviour remains coherent and dominant across performance, importance, and direction alike, most plausibly because it is defined by a hard zero-VLE-activity rule rather than a soft clustering boundary. Distinction also remains the hardest class to predict overall, because its signal depends on assessment scores that are sparse in the early-stage feature set; future work could address this with oversampling or richer pre-assessment behavioural signals.

G Summary of Results

Overall, the six experimental tiers provide a coherent set of answers to the three research questions, and the two robustness checks then show how far that answer travels beyond the setting in which it was demonstrated. The

synthesis below summarises the headline metrics and what they imply.

Baselines (B1, B1+). B1 is already strong (macro-F1 = 0.8140, AUC-ROC = 0.9085, 11 features), and B1+ improves this to 0.8273 / 0.9231 using five extra hand-crafted features without clustering. B1+ is therefore the strict benchmark for evaluating any `cluster_p` gain in I1.

RQ1 — does behavioural clustering recover meaningful learner profiles? (B2). Yes. The compact 7-feature representation separates learners better than two richer alternatives at all tested K (highest silhouette, lowest Davies–Bouldin). K = 3 is selected over statistically optimal K = 2 for interpretability, producing four actionable archetypes (High Engaged, Moderate, Disengaged, No VLE Record) with withdrawal rates from 1.3% to 95.0%. Since withdrawal is not used in clustering, this spread supports behavioural validity rather than geometric artefact.

RQ2 — does profile membership improve prediction over strong baselines? (I1). Partly, with diminishing returns as the baseline gets stronger. Adding `cluster_p` lifts macro-F1 and AUC-ROC over both B1 (+0.70pp / +0.76pp) and the richer B1+ (+0.53pp / +0.03pp), and in both cases `cluster_p` ranks third by gain importance. Neither McNemar test reaches the conventional 0.05 threshold ($p = 0.0903$ and $p = 0.1425$), so the gain is consistent in direction and magnitude across two comparisons, but remains suggestive rather than statistically conclusive.

RQ3 — does the model’s explanation differ meaningfully by learner profile? (I2). Yes. SHAP importance shifts systematically by archetype: No VLE Record receives a strongly positive `cluster_p` contribution toward Withdrawn, while the three VLE-active archetypes receive mildly negative contributions instead, and this directional pattern is stable across the B1- and B1+-based feature sets ($r = 0.797$). `cluster_p`’s overall importance rank, however, falls from 2nd of 12 features (Pair A) to 10th of 17 (Pair B) once richer hand-crafted descriptors are already present — echoing the same diminishing-returns pattern seen in RQ2 — and `num_submissions` is the one driver that ranks in the top three for every archetype.

Check 1 — does the pipeline transfer to a different module without retraining? (A1). No. Applied directly to BBB with no retraining, the FFF-trained pipeline drops from an in-domain macro-F1 = 0.802 / AUC-ROC = 0.915 to macro-F1 = 0.584 / AUC-ROC = 0.608 — a fall of 21.8 and 30.7 percentage points respectively. The larger AUC-ROC drop points to a calibration failure driven by domain shift: BBB students click far less than FFF students (median 454 versus 2,116), so cluster centroids fitted on FFF’s more active population no longer partition BBB’s engagement distribution in a behaviourally meaningful way.

Check 2 — does `cluster_p`’s advantage survive a harder, four-class task? (A2). No, on all three fronts it was designed to test. Adding `cluster_p` to B1+ turns a small macro-F1 *loss* (0.6911 $\rightarrow$ 0.6885) rather than a gain; its SHAP rank falls to 12th of 17 features and contributes essentially zero mean |SHAP| to the Distinction and Pass classes; and its directional coherence from RQ3 holds cleanly only for the No VLE Record archetype, whose behaviour is governed by a hard zero-VLE-activity rule rather than a soft clustering boundary.

Taken together, RQ1–RQ3 are supported in the single-module binary setting of Tiers B1–I2: clustering yields meaningful profiles, adds modest but consistent directional predictive improvement, and supports profile-specific explanations. The robustness checks qualify this result: A1 fails to transfer across modules without retraining, and A2 shows the advantage weakens in the harder four-class task. Across all tiers, No VLE Record remains the most stable and informative archetype.

V CONCLUSION

This project set out to test whether cluster-derived behavioural profiles can support earlier and more interpretable at-risk prediction than raw-feature baselines alone. Across six experimental tiers, the answer is a qualified yes: VLE clickstream data can be converted into risk-differentiated learner archetypes that add a small but consistent directional predictive improvement over strong baselines and reveal explanation patterns a population-level model would hide; two robustness checks, however, show this benefit is narrower than Tiers B1–I2 alone would suggest.

Contributions. Section E identified three linked gaps: profiling work often stops before prediction, prediction work often assumes a homogeneous learner population, and explainability is usually reported only at population level. This project addresses them with one reproducible OULAD pipeline that (i) derives four behavioural archetypes from a compact, justified feature set (RQ1), (ii) tests incremental predictive value of profile membership against two matched baselines (RQ2), (iii) shows that SHAP explanations differ systematically by archetype (RQ3), and (iv) adds explicit cross-module transfer and multi-class robustness checks instead

of reporting B1–I2 alone.

RQ1. A compact, seven-feature description of engagement volume and regularity separates OULAD learners more cleanly than two richer alternatives, achieving the highest silhouette score and lowest Davies–Bouldin Index at every tested K . $K = 3$ is chosen over the statistically optimal but educationally crude $K = 2$, yielding four archetypes — High Engaged, Moderate, Disengaged, and No VLE Record — whose withdrawal rates diverge sharply (1.3% to 95.0%). This composition is fitted on FFF alone and, as Tier A1 shows, does not carry over unchanged to BBB.

RQ2. Adding `cluster_p` to the compact B1 baseline raises macro-F1 by 0.0070 and AUC-ROC by 0.0076; against the stronger, hand-engineered B1+ baseline the gain shrinks to 0.0053 and 0.0003. `cluster_p` ranks third by importance in both models, so the effect is not tied to one feature combination, but neither McNemar test reaches the 0.05 significance threshold ($p = 0.0903, 0.1425$), so the gain is consistent in direction rather than statistically confirmed.

RQ3. `cluster_p`’s SHAP contribution is strongly positive for No VLE Record and mildly negative for the three VLE-active archetypes, a direction stable across both feature sets ($r = 0.797$) even as its overall rank falls from 2nd to 10th once richer features are present. `num_submissions` is the top driver in every archetype, but Disengaged learners are instead driven primarily by `mean_score` and `unique_resources` — consistent with High Engaged and No VLE Record being the pair with weakest Kendall τ rank agreement.

Limitations. Tier A1 shows the FFF-trained pipeline does not transfer to BBB without retraining (macro-F1 -21.8 pp, AUC-ROC -30.7 pp), traced to BBB’s substantially lower median VLE activity, so the B2 profiles are best read as module-specific rather than universal. Tier A2 shows `cluster_p`’s advantages do not survive a harder four-class task: its macro-F1 contribution turns slightly negative, its SHAP rank drops to 12th of 17, and its directional coherence holds cleanly only for No VLE Record — the one profile defined by a hard rule rather than a soft boundary. Finally, SHAP values measure association rather than causation, and the `cluster_p` label used throughout I1, I2, A1, and A2 was deliberately drawn from the weaker B2+ partition rather than the better-separated compact B2 solution, to keep downstream feature sets consistent.

Practical implications. For deployment, cluster-derived profiles should supplement, not replace, raw-feature models. The No VLE Record group is the most reliable intervention target because its predictive value, importance, and SHAP direction stay stable across tiers and checks; the softer K-means archetypes should be used as lower-confidence secondary signals. A1 also shows that clustering and scaling must be re-fitted when moving to a new module, because transferring FFF centroids to BBB causes calibration failure. Finally, if socioeconomic predictors are reintroduced, the fairness check highlighted by AlFaress et al. (2025) should be repeated.

Future work. Three extensions follow: (1) re-fit centroids and scaler per target module, combined with the A1 calibration/threshold workflow, to mitigate domain shift; (2) replace unordered integer `cluster_p` with ordinal or probability-weighted encoding to better exploit archetype order; and (3) derive clustering directly for multi-class outcomes and validate on more modules beyond FFF/BBB.

Closing remark. Cluster-derived profiles are a suggestive but modest source of predictive and explanatory value within a single, previously-seen module; neither their predictive edge nor their explanatory pattern should be assumed to generalise across modules or under a harder task without direct testing. This qualified result nonetheless fills the gap identified in the literature: to the best of our knowledge based on the reviewed literature, no prior OULAD study had combined behavioural clustering, predictive evaluation, and cluster-specific explanation alongside an explicit transfer and robustness check, and reporting both the gain and its limits honestly is, in itself, a more useful contribution to practitioners than a headline accuracy number alone.

GENERATIVE AI USE DECLARATION

I used Claude 5.0 as a generative AI assistant for language support (grammar checking, wording refinement, and simplification), limited literature search support, and minor code completion. I did not use generative AI to generate the core research ideas, experimental design, results, or final conclusions. All substantive academic content is my own original work, and I have reviewed and take full responsibility for the final submission.

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